ACTUATOR POSITION SENSOR (TURBOCHARGER WITH VARIABLE TURBINE GEOMETRY) [SKYACTIV-D 1.5]

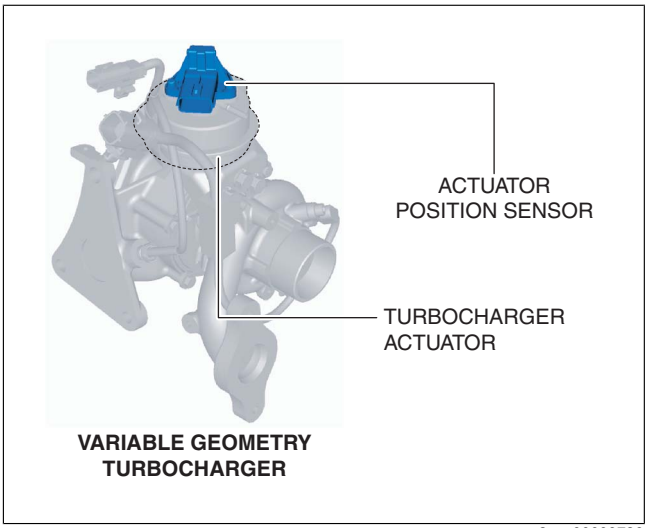
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Purpose/Function

- The actuator position sensor detects the open/closed condition of the variable nozzle which drives the turbocharger actuator of the variable geometry turbocharger as basic information for mainly boost control.
- The actuator position sensor detects the open/closed condition of the valve which drives the turbocharger actuator of the variable geometry turbocharger and sends the signal to the PCM as a turbocharger actuator opening angle signal.

Construction

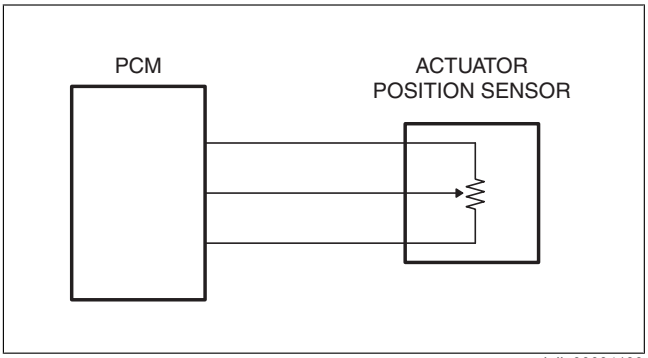
- The actuator position sensor of the variable geometry turbocharger is located on the turbocharger actuator.
- The actuator position sensor of the variable geometry turbocharger has adopted a Hall element to the detection area for detecting the open/closed condition of the valve which drives the turbocharger actuator of the variable geometry turbocharger.



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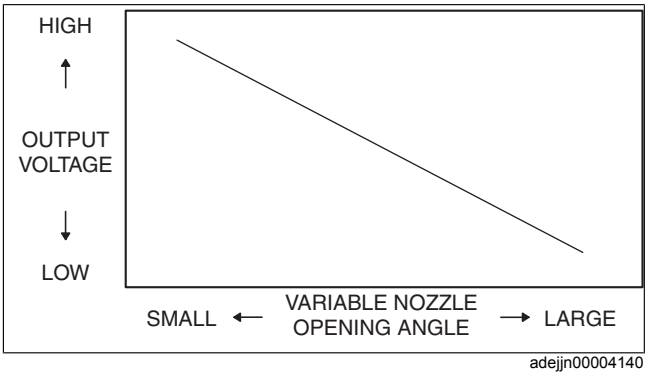
Operation

- The signal, which changes according to the open/closed condition of the variable nozzle driving the turbo actuator of the variable geometry turbocharger, is sent to the PCM.



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Actuator position sensor (turbocharger with variable turbine geometry) characteristic



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Fail-safe

DTC No.	Fail-safe function
P2263:00	• Limits the engine torque. • Limits the upper limit of the engine speed. • Inhibits the EGR control. • Inhibits the boost control. • Inhibits the auto DPF regeneration control. • Inhibits the compulsory DPF regeneration control. • PCM restricts engine-transaxle integration control. (ATX)
P2564:00	• Limits the engine torque. • Inhibits the EGR control. • Inhibits the boost control. • Inhibits the auto DPF regeneration control. • Inhibits the compulsory DPF regeneration control. • Inhibits engine-stop by operating the i-stop function. • PCM restricts engine-transaxle integration control. (ATX)
P2565:00	